

## CULVERTCRAWL · REQUIREMENTS

# CulvertCrawl requirements

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## DOCUMENT CONTROL

|             |                                            |         |                |
|-------------|--------------------------------------------|---------|----------------|
| Document ID | CVC-REQ-001                                | Version | 0.2            |
| Status      | <b>Draft</b>                               | Date    | 2026-09-25     |
| Author      | Amish Chadha                               | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 2</b> Technology concept formulated |         |                |
| Repository  | github.com/BoujeeEnjinia1701/culvertcrawl  |         |                |
| Source      | docs/03-requirements.md @ ff871fe          |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                               |
|---------|------------|--------------|--------------------------------------------------------------------------------------|
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with first-order status against the concept |
| 0.1     | 2026-09-24 | Amish Chadha | Initial scaffold                                                                     |

## CulvertCrawl requirements

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These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users (see CVC-PRB-001), and will be checked by calculation at TRL 3. The status column gives the first-order result from the design precis (CVC-PRC-001); every status is an estimate.

The **design case** used throughout is a straight 600 mm (24 in) corrugated steel culvert, 50 m long, on a 5 % slope, with a wet silt invert and up to 150 mm of water flowing at 0.5 m/s.

■ Requirements

| ID  | Requirement                 | Target                                                                                                                                                                                        | Verification (TRL 3 or later)                         | Status (estimate)                                                                                 |
|-----|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| R1  | Fit the target pipes        | Drive in round pipes of 300 to 900 mm (12 to 36 in) inside diameter; crawler no wider than 180 mm and no taller than 140 mm                                                                   | Model check in 300 mm and 900 mm pipe sections        | Met by design (170 x 125 mm)                                                                      |
| R2  | Reach                       | Drive 50 m into the design case pipe and back, pulling the tether, with 20 % traction reserve                                                                                                 | Traction and tether drag calculation; later pull test | Met, margin thin (about 56 m at 20 % reserve)                                                     |
| R3  | Recover without entry       | Crawler can be pulled out by the tether if disabled; tether break strength 1 kN or more, at least 5 times the worst stuck-pull estimate                                                       | Tether datasheet; pull calculation                    | Met by specification                                                                              |
| R4  | Video                       | 1080p video at 25 frames per second or more, live at the operator and recorded, with distance and time overlay; lighting to show the far wall of a 900 mm pipe                                | Camera and link budget; later image review            | Met on paper; lighting in 900 mm pipe unverified                                                  |
| R5  | Cross-section profile       | Measure the wall profile in the ring plane and report mean diameter and ovality (deflection) to within 1 % of nominal diameter, every 0.1 m of travel or less                                 | Error budget; later calibration pipes                 | Met at 300 to 600 mm; <b>at risk at 900 mm</b> (about 1 % or worse at the top of the pipe)        |
| R6  | Locate observations         | Distance from the pipe mouth within 1 % or 0.2 m, whichever is larger; pitch and roll within 1 degree                                                                                         | Encoder and IMU datasheets                            | Met on paper                                                                                      |
| R7  | Drive in water and silt     | Operate in 150 mm of water flowing at 0.5 m/s over a silt invert; survive full submersion to 1 m for 30 min (IP68 target)                                                                     | Drag and buoyancy calculation; later tank test        | Driving met on paper; <b>IP68 unverified</b> ; traction when fully submerged is about half of dry |
| R8  | Climb                       | Hold position on a 5 % slope with power off; cross a 40 mm step or joint offset                                                                                                               | Worm gear self-locking; track geometry                | Hold met by design; step crossing unverified                                                      |
| R9  | Endurance                   | 4 h or more of mixed driving and profiling per charge                                                                                                                                         | Energy budget                                         | Met (about 6 h)                                                                                   |
| R10 | Portable and quick          | Whole kit 25 kg or less, no single case over 10 kg, carried by two people in one trip; set up from vehicle to driving in 10 min or less                                                       | Mass estimate; setup sequence                         | Met on mass (about 18 kg); setup time unverified                                                  |
| R11 | Electrical and laser safety | Tether and crawler 48 V DC nominal (below the 60 V DC extra-low-voltage limit), fused, with overcurrent cutoff and an emergency stop at the surface; laser Class 2 or lower under IEC 60825-1 | Design review                                         | Met by design                                                                                     |
| R12 | Affordable                  | Full kit (crawler, 60 m tether, reel, surface box) \$800 or less in parts, operator laptop excluded                                                                                           | Priced BOM ( bom/ bom.csv )                           | <b>Not met</b> (about \$900, about 13 % over)                                                     |
| R13 | Open and repairable         | Common parts; every tether and module joint pluggable; open file formats (MP4 video, CSV                                                                                                      | Design review                                         | Met by design                                                                                     |

| ID | Requirement | Target                                                | Verification (TRL 3 or later) | Status (estimate) |
|----|-------------|-------------------------------------------------------|-------------------------------|-------------------|
|    |             | profiles, JSON observation log with PACP-style codes) |                               |                   |

## ■ Requirements not met or at risk

- **R12 (cost) is not met.** The first priced BOM is about \$900 against \$800. Options are in `docs/REVIEW.md` ; the budget change is proposed, awaiting Amish.
- **R5 is at risk at 900 mm.** With the laser ring 200 mm ahead of the lens, the top of a 900 mm pipe is seen about 76 degrees off the camera axis, at the edge of a fisheye lens, where 1 % is marginal.
- **R7 submersion is unverified.** IP68 to 1 m is a target for a hull with rotating shaft seals; seal leakage is the most likely failure. Laser profiling does not work below the water surface (refraction), so the profile covers only the wall above the water line.
- **R2 margin is thin.** Reach depends on tether drag; a heavier tether, bends or a corrugated invert could cut reach below 50 m.

## ■ Assumptions

- Track-to-silt friction coefficient about 0.6 and track motion resistance about 0.15 of weight; tether-to-pipe friction coefficient about 0.5.
- Crawler mass about 5.5 kg with ballast; tether about 55 g/m; both are estimates until parts are selected.
- 50 m reach covers most road crossings; 60 m of tether allows 10 m from the reel to the mouth.
- A 1 % profile accuracy resolves the roughly 5 % deflection limit used for plastic pipe acceptance with a 5 to 1 margin.
- The operator's own laptop runs the viewing and profiling software.